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CSI-Feedback-Driven Link Adaptation in 5G NR Downlink MIMO: A Link- Level Simulation Study

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I hereby declare that, except where specifically indicated, the work submitted herein is my own original work.

Signed Kaiyuan Bao 包铠源 date 30 May 2026

Acknowledgement

Technical Abstract

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Background and Motivation

Methodology

Key Results and Interpretation

Conclusion

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List of Abbreviations

Abbreviation	Definition
5G	Fifth Generation
BLER	Block Error Rate
CDL	Clustered Delay Line
CP	Cyclic Prefix
CQI	Channel Quality Indicator
CSI	Channel State Information
CSI-RS	CSI Reference Signal
DL	Downlink
DM-RS	Demodulation Reference Signal
FR	Frequency Range
gNB	Next Generation Node B
HARQ	Hybrid Automatic Repeat Request
LDPC	Low-Density Parity-Check
LOS	Line-of-Sight
MCS	Modulation and Coding Scheme
MIMO	Multiple-Input Multiple-Output
NLOS	Non-Line-of-Sight
NR	New Radio
OFDM	Orthogonal Frequency Division Multiplexing
PDCCH	Physical Downlink Control Channel
PDSCH	Physical Downlink Shared Channel
PMI	Precoding Matrix Indicator
QAM	Quadrature Amplitude Modulation
QPSK	Quadrature Phase Shift Keying
RB	Resource Block
RE	Resource Element
RI	Rank Indicator
SCS	Subcarrier Spacing
SINR	Signal-to-Interference-and-Noise Ratio
SISO	Single-Input Single-Output
SNR	Signal-to-Noise Ratio
TDL	Tapped Delay Line
UE	User Equipment
UL	Uplink

1 Introduction

1. 5G NR, Physical Layer, MIMO background. Why 5G needs mimo and adaptive transmission
2. Importance of CSI
3. Why fixed-rank, fixed-modulation BER curve is not enough 3. Matlab 5G toolbox simulation
4. Baseline scenario
5. RQs
6. Structure of my research

2 Background and Methodology

2.1 5G NR Downlink Physical Layer Overview

1. OFDM
2. PDSCH (DL)
3. DM-RS
4. CSI-RS
5. Slot, RB
6. Definition of BER/BLER/Throughput
7. Why throughput not BER

2.2 MIMO Transmission and CSI Feedback

Specifics of CSI: PMI, RI, CQI

Feedback loops

How the modulation scheme/TCR/rank changes based on the feedback

2.3 CDL Channel Model and Doppler Shift

CDL model definition

CDL ABCDE difference, why CDL-C default

Doppler shift definition and impact

2.4 MATLAB Simulation Framework

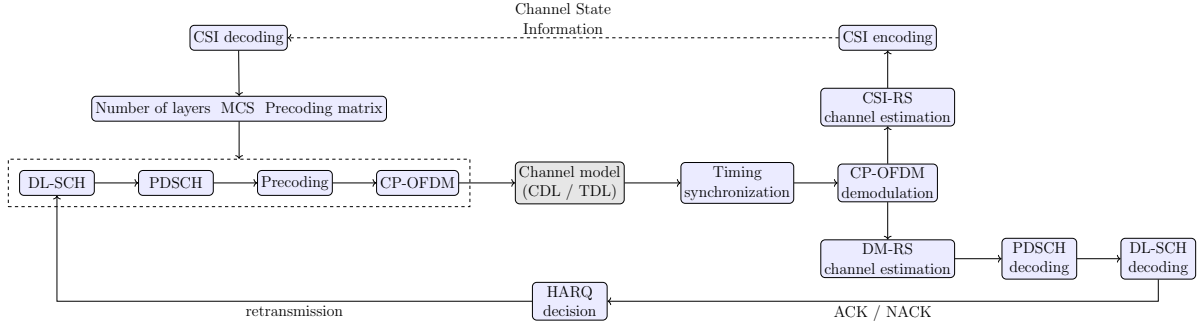


Figure 1: Overview of the CSI-based downlink transmission and feedback framework, adapted from [1]

Table 2: System and channel configuration used in the simulation

Parameter	Value
Carrier frequency	FR1, 3.5 GHz
Subcarrier spacing	30 kHz
Channel bandwidth	106 RBs
Antenna configuration	4 Tx, 4 Rx
Maximum number of layers	4
Channel model	CDL-C
Mobility scenarios	Low (10 Hz), Medium (100 Hz), High (390 Hz)
Simulation type	Downlink link-level

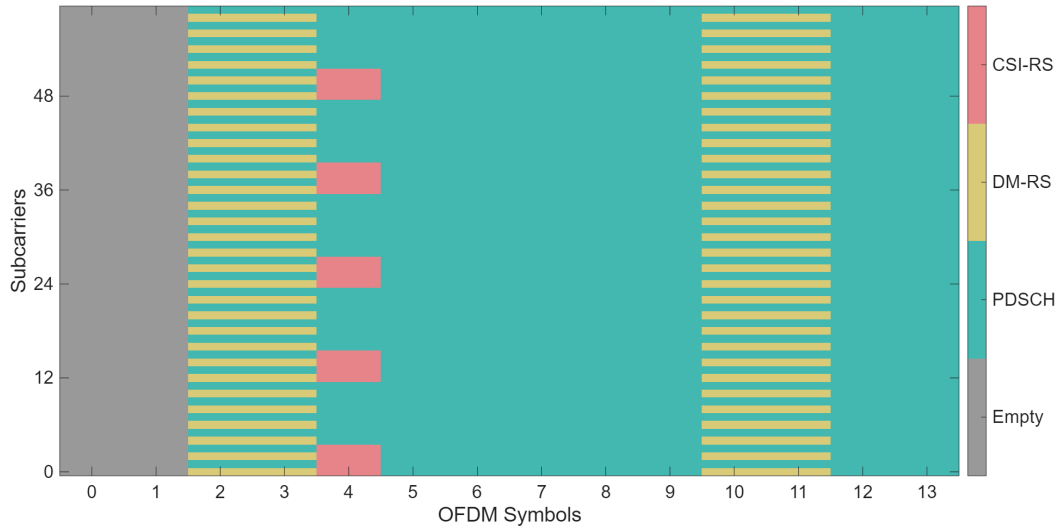


Figure 2

2.5 System Configuration and Experimental Design

Table 3: Baseline CSI feedback configuration

Parameter	Baseline setting
CSI report periodicity	10 slots
CQI reporting mode	Wideband
PMI reporting mode	Wideband
Subband size	16 RBs
RI configuration	Adaptive
Codebook type	Type I
Codebook mode	Mode 1

In addition to baseline setting, further tests will be done by changing....

Table 4: Simulation studies and purpose

Study Group	Purpose	No. of simulations
Baseline	CDL-C 4x4 reference	1
Fixed-rank controls	Isolate RI effect	
Antenna sweep	MIMO robustness	
CDL sweep	Channel robustness	
Doppler sweep	Mobility effect	3
CSI period sweep	CSI freshness	

3 Baseline CSI-Feedback Downlink Behaviour

3.1 Baseline CDL-C 4x4 Throughput Behaviour

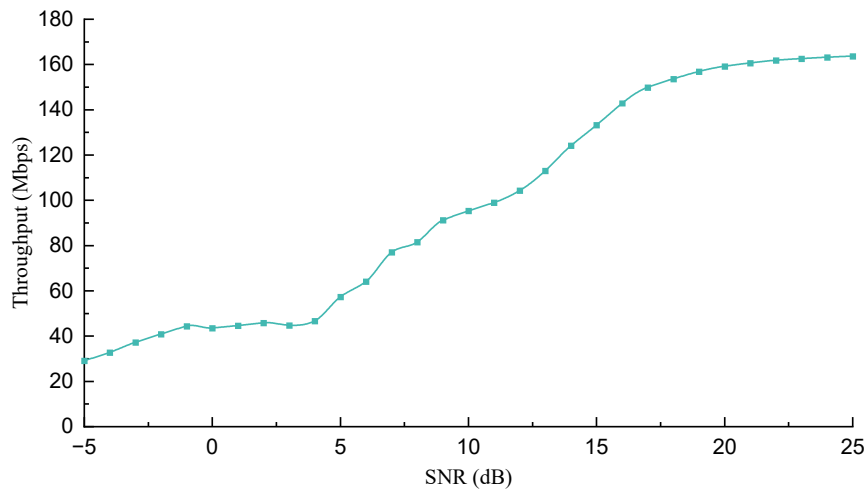


Figure 3: Baseline throughput performance of the CDL-C 4x4 downlink configuration with adaptive CSI feedback

3.2 CSI-driven Link Adaptation Statistics

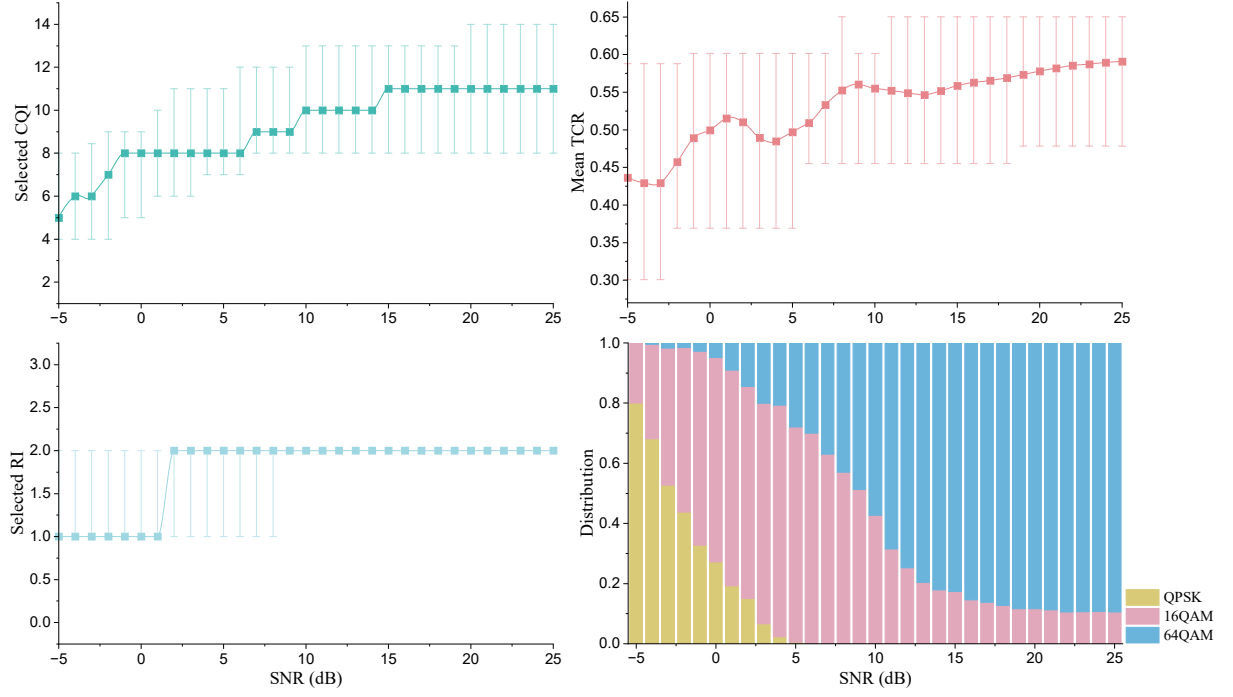


Figure 4

3.3 Fixed-Rank Controls

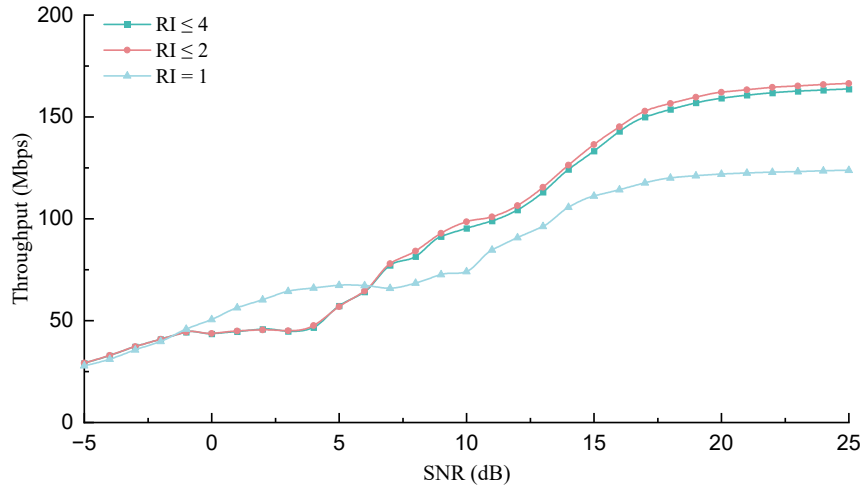


Figure 5: Throughput performance under different maximum layer configurations

* To show how the adaptive layer (RI) actually performs

3.4 Summary of Baseline Findings

1. The above result is the baseline 2. Examines the impact of CSI and how it actually works

4 Robustness across MIMO and Channel Configurations

*** The baseline is then tested against changes in antenna configuration and CDL channel profile, showing which behaviours are robust and which depend on spatial/channel conditions. [Should not go too deep into this section]

This chapter examines whether the baseline behaviour observed in the CDL-C 4x4 case remains interpretable under different antenna configurations and CDL channel profiles.

4.1 Antenna Configuration Sweep

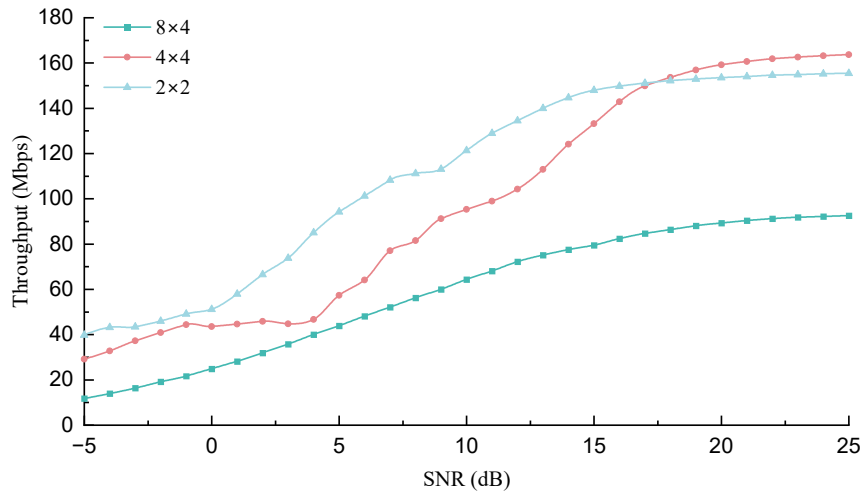


Figure 6: Still need to check 1x1 and 4x8 results

4.2 CDL Channel Profile Sweep

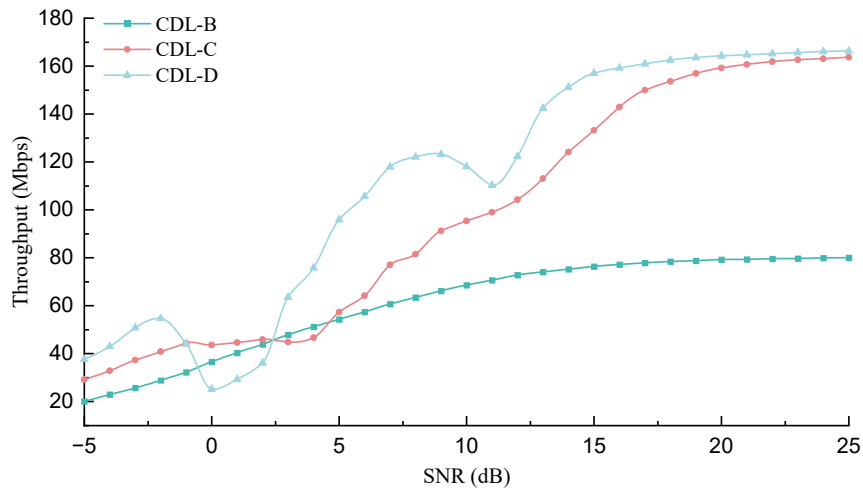


Figure 7

4.3 One-layer Control Experiments

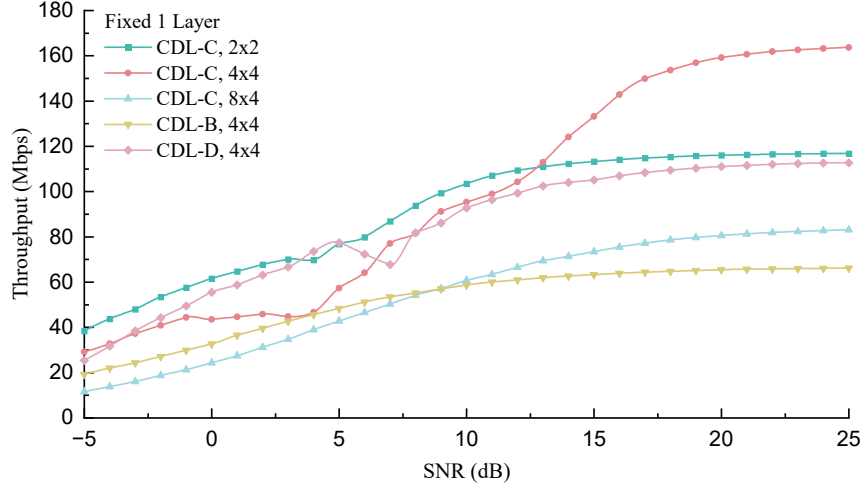


Figure 8: Some issue with [2x2] and [8x4] curves, re-simulate

5 Impact of Mobility and CSI Reporting Periodicity

5.1 Doppler Shift under Default CSI Reporting Period

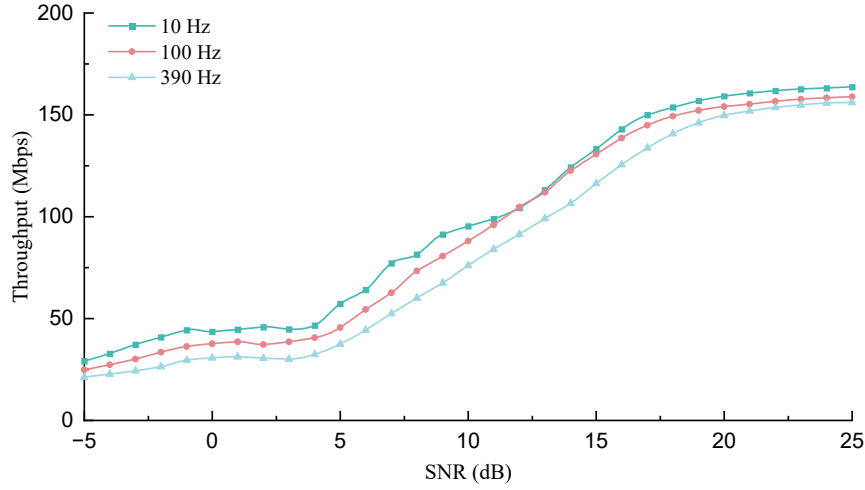


Figure 9

Table 5: Interpolated SNR values for different target throughput levels

Target Throughput (Mbps)	SNR 10 Hz (dB)	SNR 100 Hz (dB)	SNR 390 Hz (dB)
50	4.31	5.50	6.69
80	7.66	8.91	10.50
110	12.66	12.73	14.35
140	15.70	16.23	17.89

[†] SNR values obtained using linear interpolation.

5.2 Reporting Periodicity at Low and High mobility

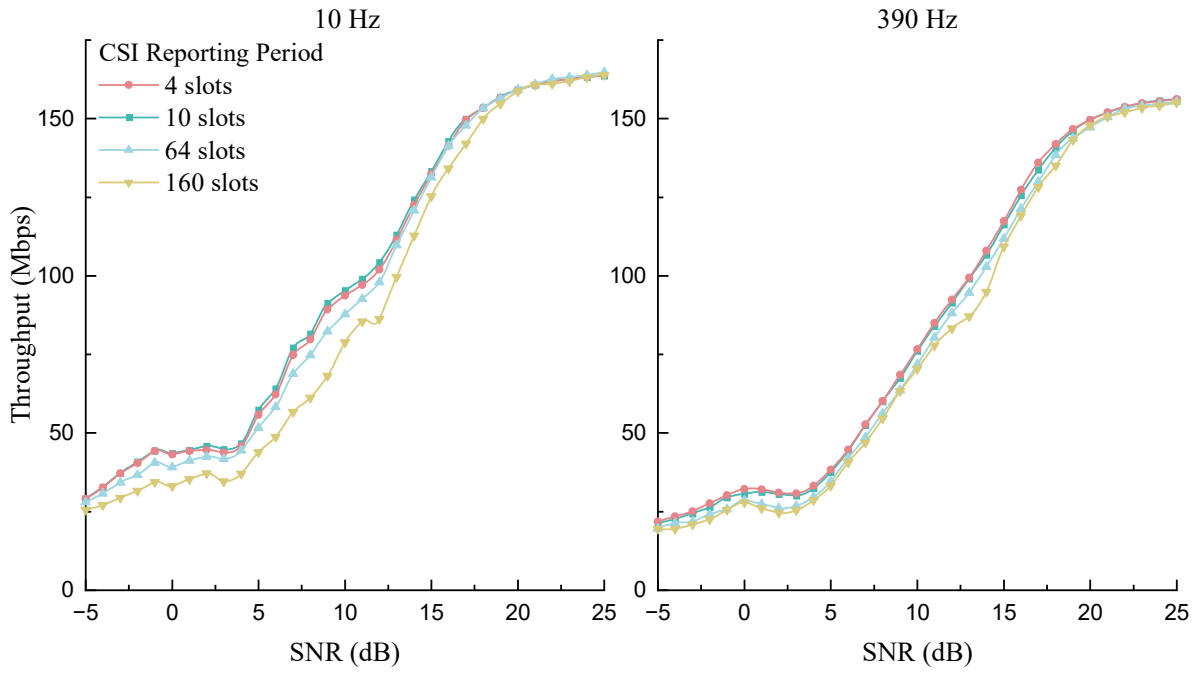


Figure 10

$$\Delta\gamma_{f_D,P} = \gamma_{f_D,P}(0.5R_{f_D,[10,0]}^{\max}) - \gamma_{f_D,[10,0]}(0.5R_{f_D,[10,0]}^{\max}). \quad (1)$$

Table 6: SNR required to reach 50% of the reference peak throughput of the [10, 0] baseline, and the corresponding SNR offset relative to this baseline. Positive values indicate an SNR penalty, while negative values indicate an SNR gain

Doppler	CSI period	$\gamma_{50\%}$	$\Delta\gamma$
10 Hz	4 slots	8.23 dB	0.18 dB
10 Hz	10 slots	8.05 dB	–
10 Hz	64 slots	8.95 dB	0.90 dB
10 Hz	160 slots	10.47 dB	2.42 dB
100 Hz	4 slots	8.48 dB	-0.50 dB
100 Hz	10 slots	8.98 dB	–
100 Hz	64 slots	11.01 dB	2.03 dB
100 Hz	160 slots	10.76 dB	1.78 dB
390 Hz	4 slots	10.28 dB	-0.09 dB
390 Hz	10 slots	10.37 dB	–
390 Hz	64 slots	10.84 dB	0.47 dB
390 Hz	160 slots	11.21 dB	0.84 dB

[†]: 10 Hz: 163.89Mbps; 100 Hz: 161.01 Mbps; 390 Hz: 157.98 Mbps.

No need for 2 digit precision, will change later

5.3 Joint Interpretation: CSI Freshness

Summarize 4.2 and 4.3 CSI feedback should not be evaluated only by its granularity or reporting frequency. Its value depends on whether the reported channel state remains relevant over the time scale of channel variation.

5.4 Discussion of Feedback Overhead and Practical Trade-Off

1. What exactly is the cost of CSI feedback 2. Under what scenarios are frequent CSI report actually useful

6 Discussion and Conclusion

6.1 Interpretation of Results

6.2 Limitations

6.3 Future Work

7 References

- [1] MathWorks, “NR PDSCH Throughput Using Channel State Information Feedback,” 5G Toolbox Documentation. [Online]. Available: <https://www.mathworks.com/help/5g/ug/nr-pdsch-throughput-using-csi-feedback.html>

8 Appendix